

1 minute Review:

vector operations

mag $|v| = \sqrt{\sum v_i^2}$

Scalar mult $c v = \langle c v_1, \dots, c v_n \rangle$

vector add $v + w = \langle v_1 + w_1, \dots, v_n + w_n \rangle$

→ cross-fading

Convex combinations $(1-t)p + tq$

odds & Evils

$|x - y|$ is the distance formula



take the magnitude of the distance.

• special basis vectors in $\mathbb{R}^3$

$$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\langle a, b, c \rangle = a i + b j + c k$$

The dot Product

Def: $v = \langle v_1, \dots, v_n \rangle$ & $w = \langle w_1, \dots, w_n \rangle$ The dot product is

$$v \cdot w = \sum_{i=1}^n v_i w_i$$

Ex: $\langle 4, 2, -1 \rangle \cdot \langle -2, 6, 0 \rangle = -8 + 12 + 0 = 4$

$$\begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} -8 \\ +12 \\ 0 \end{bmatrix} = 4$$

$$i \cdot i = j \cdot j = k \cdot k = 1$$

why $i \cdot j = j \cdot k = k \cdot i = 0$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Properties of dot Product

• Positive definite

$$v \cdot v = |v|^2$$

• Symmetric

$$v \cdot w = w \cdot v$$

• bilinear

why you can distribute

$$u \cdot (cv + dw) = cu \cdot v + du \cdot w$$

$$(i+j) \cdot (j-k)$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 1$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

or

$$(i+j) \cdot (j-k) = 1$$

why dot Product?

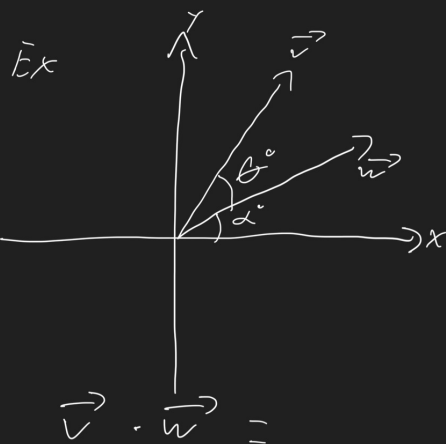
when is this large?

$$v \cdot w = v_1 w_1 + v_2 w_2 + v_3 w_3 + v_4 w_4$$

1. dot Product only cares about relative position not absolute position.

Look into how to compute this for 3 dimensions and higher.

Review this Portion again.



$$\vec{v} = (|\vec{v}| \cos(\theta + \alpha), |\vec{v}| \sin(\theta + \alpha))$$

$$\vec{w} = (|\vec{w}| \cos \alpha, |\vec{w}| \sin \alpha)$$

$$\vec{v} \cdot \vec{w} = |\vec{v}| |\vec{w}| (\cos \alpha \cos(\theta + \alpha) + \sin \alpha \sin(\theta + \alpha))$$

$$= |\vec{v}| |\vec{w}| (\cos \alpha \cos(\theta + \alpha) - \sin \alpha \sin(\theta + \alpha))$$

$$= |\vec{v}| |\vec{w}| \cos(\theta + \alpha + \alpha)$$

$$|\vec{v}| |\vec{w}| = |\vec{v}| |\vec{w}| \cos \theta \rightarrow \text{Geometric Formula}$$

Cosine angle summation formula

they are perpendicular if they = 0

The max two vectors can get is when they line up

Formula

They can also show us it

